

Instructions

- Answer all eight questions. Show sufficient working for method marks.
- A calculator may be used unless a question states otherwise.
- Write clearly in the spaces provided. Additional pages may be attached.

1

Evaluate the definite integral, showing the antiderivative you use.

[5 marks]

$$\int_0^2 (3x^2 - 4x + 1) dx$$

2

Find the equation of the tangent to the curve below at the point where $x = 2$.

[5 marks]

$$y = x^3 - 2x$$

3

Solve the differential equation given that $y(0)=2$.

[6 marks]

$$\frac{dy}{dx} = 3y$$

4

Decompose into partial fractions, stating the form you assume before finding the coefficients.

[6 marks]

$$\frac{5x + 1}{(x - 1)(x + 2)}$$

5

Integrate by parts, stating your choice of u and dv .

[5 marks]

$$\int x e^x dx$$

6

Determine whether the improper integral converges, and evaluate it if it does.

[5 marks]

$$\int_1^{\infty} \frac{1}{x^2} dx$$

7

Find the stationary points and classify each using the second derivative.

[6 marks]

$$f(x) = x^3 - 3x^2 + 4$$

8

Use the chain rule to differentiate, then evaluate at $x=0$.

[5 marks]

$$y = \sin(2x^2 + x)$$
